

TE_pangenome_construction workflow

TE_pangenome/scripts/TE_pangenome_construction/

Key:

Process

script

inputs

outputs

Sample download and alignment

SRA Download	sra_download_bulk.sbatch	
	data/ sra_HGSVC3.txt	outputs/ TE_pangenome_construction/ fastqs/ file.fastq
Gzip files	gzip.sbatch	
	outputs/fastqs/ file.fastq	outputs/ TE_pangenome_construction/fastqs/ file.fastq.gz
NanoStat	nanoStat_fastqs.sbatch	
	outputs/fastqs/ file.fastq.gz	outputs/ TE_pangenome_construction/NanoStat_fastqs/ file_stats.txt
NanoPlot	nanoPlot_summary_fastqs.sbatch	
	outputs/fastqs/ file.fastq.gz	outputs/ TE_pangenome_construction/NanoPlot_fastqs/ file_stats.txt
Align reads	minimapAlign.sbatch	
	data/ hg38.main.fa	outputs/ TE_pangenome_construction/bams_bySRA/ file.bam file.sorted.bam file.sorted.bam.bai
	outputs/fastqs/ file.fastq.gz	
Merge bams into sample IDs	merge_bams.py	
	data/ sample_ID_bam_list.csv	outputs/ TE_pangenome_construction/bams_merged/ file.merged.bam
	outputs/bams_bySRA/ file.sorted.bam file.sorted.bam.bai	
Sort & index merged bams	sort_index_mergedbams.sbatch	
	outputs/bams_merged/ file.merged.bam	outputs/ TE_pangenome_construction/bams_merged_sorted/ file.merged_sorted.bam file.merged_sorted.bam.bai
NanoStat	nanoStat_bams.sbatch	
	outputs/bams_merged_sorted / file.merged_sorted.bam file.merged_sorted.bam.bai	outputs/ TE_pangenome_construction/NanoStat_bams/ file_stats.txt
NanoPlot	nanoPlot_summary_bams.sbatch	
	outputs/NanoStat_bams/ file_stats.txt	outputs/ TE_pangenome_construction/NanoPlot_bams/ file_stats.txt

Pangenome Construction

Sniffles2	sniffles.sbatch		
	data/ hg38.main.fa	outputs/ TE_pangenome_construction/sniffles/ file.snf file.vcf	
	outputs/bams_merged_sorted/ file.merged_sorted.bam file.merged_sorted.bam.bai		
Sniffles2 Population	Sniffles_popn.sbatch		
	data/ hg38.main.fa	outputs/ TE_pangenome_construction/sniffles/ genotypes_unfiltered.vcf sniffles2_variants.vcf	
	outputs/sniffles/ file.snf		
RepeatMasker	repmask_vcf.sbatch		
	outputs/sniffles_popn/ sniffles2_variants.vcf	outputs/ TE_pangenome_construction/repeatmasker/ indels.fa	
Build, map & filter repeats	build_run_Onecode.sbatch		
	data/ build_dictionary.pl	outputs/ TE_pangenome_construction/repeatmasker/ indels.fa.out OneCode_LTR.dic file.transposons.csv file.copynumber.csv file.elem_sorted.csv file.ltr.csv	
	outputs/repeatmasker/ indels.fa		
	filterRepeats_Onecode.sbatch		
Annotate repeats	outputs/repeatmasker/ indels.fa	outputs/ TE_pangenome_construction/repeatmasker/ indels.fa.onecode.out ALL.elem_sorted.csv	
	annotate_vcf_updated.R		
	outputs/repeatmasker/ sniffles2_variants.vcf indels.fa.onecode.out	outputs/ TE_pangenome_construction/repeatmasker/ vcf_annotation	
	annotateRepeats_Onecode.sbatch	outputs/ TE_pangenome_construction/repeatmasker/ indels.fa.fai indels.fa.masked indels.length span_no_NA Twp.txt	SVA_VNTR.bed SVA_VNTR.txt hdr_file vcf_annotation_1.vcf.gz genotypes.sorted.vcf genotypes.remasked.vcf.gz
Filter for < 3 TE families per variant	TE_family_filter3.sbatch		
	outputs/repeatmasker/ genotypes.remasked.vcf.gz	outputs/ TE_pangenome_construction/filtered_pangenome/ pangenome3TE.vcf	
Filter for Target Site Duplications (TSDs)	run_fix_vcf.py → fix_vcf.py		
	data/ hg38.main.fa	outputs/ TE_pangenome_construction/filtered_pangenome/ pangenome3TE_TSDfilt.vcf	
	outputs/filtered_pangenome/ pangenome3TE.vcf		
	tsd_prep.sbatch		
	data/ hg38.main.fa	outputs/ TE_pangenome_construction/filtered_pangenome/TSD/ pangenome3TE_TSDfilt_prep.vcf gLength.txt oneHit_SV_coordinates.bed oneHit_SV_coordinates_win.bed flanking_sequences.fasta oneHit_indels.fa.masked SV_sequences_L_R_trimmed_WIN.fa indels.txt	
	outputs/filtered_pangenome/ pangenome3TE_TSDfilt.vcf		
	outputs/repeatmasker/ indels.fa.masked		
	tsd_search.sbatch		
	outputs/repeatmasker/ genotypes.remasked.vcf.gz	outputs/ TE_pangenome_construction/filtered_pangenome/TSD/ L.fasta R.fasta file.TSD_full_log.txt	
	tsd_report.sbatch		
	outputs/repeatmasker/ genotypes.remasked.vcf.gz	outputs/ TE_pangenome_construction/filtered_pangenome/TSD/ indels.txt.TSD_summary.txt indels.txt.TSD_full_log.txt	
	outputs/filtered_pangenome/TSD/ indels.fa.masked	outputs/ TE_pangenome_construction/filtered_pangenome/ TE_pangenome_21.vcf TE_pangenome_21.vcf.gz TE_pangenome_21.vcf.gz.tbi	

RNA_processing

TE_pangenome/scripts/RNA_Processing

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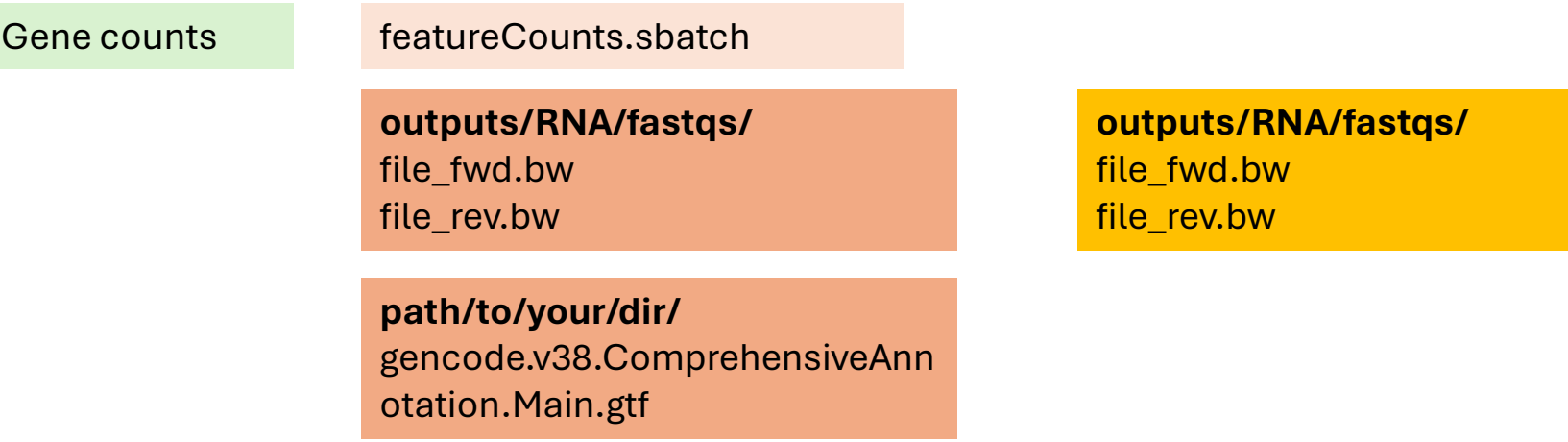
RNA processing



FeatureCounts

TE_pangenome/scripts/featureCounts

Gene counts



Differential Expression

TE_pangenome/scripts/DESeq2

Differential Expression

